

Data Structure and Algorithms

Exercise 2: E-commerce Platform Search Function

CODE :

```
class Product {
    int productId;
    String productName;
    String category;
    public Product(int productId, String productName, String category) {
        this.productId = productId;
        this.productName = productName;
        this.category = category;
    }public void display() {
        System.out.println("Product ID: " + productId +
            ", Name: " + productName +
            ", Category: " + category);
    }
}

public class Main {
    //Linear Search
    public static Product linearSearch(Product[] products, int targetId) {
        for (Product product : products) {
            if (product.productId == targetId) {
                return product;
            }
        }
        return null;
    }
}
```

```
}
```

//Binary Search

```
public static Product binarySearch(Product[] products, int targetId) {  
    int low = 0;  
    int high = products.length - 1;  
    while (low <= high) {  
        int mid = (low + high) / 2;  
        if (products[mid].productId == targetId) {  
            return products[mid];  
        }  
        else if (products[mid].productId < targetId) {  
            low = mid + 1;  
        }  
        else {  
            high = mid - 1;  
        }  
    }  
    return null;  
}  
  
public static void main(String[] args) {  
    Product[] products = {  
        new Product(101, "Laptop", "Electronics"),  
        new Product(102, "Mobile", "Electronics"),  
        new Product(103, "Shoes", "Fashion"),  
        new Product(104, "Watch", "Accessories"),  
        new Product(105, "Book", "Education")  
    };  
};
```

```
int searchId = 104;
System.out.println("Linear Search Result:");
Product result1 = linearSearch(products, searchId);
if (result1 != null)
    result1.display();
else
    System.out.println("Product not found");
System.out.println("\nBinary Search Result:");
Product result2 = binarySearch(products, searchId);
if (result2 != null)
    result2.display();
else
    System.out.println("Product not found");
}
}
```

OUT PUT:

Output

```
Linear Search Result:
Product ID: 104, Name: Watch, Category: Accessories

Binary Search Result:
Product ID: 104, Name: Watch, Category: Accessories

=== Code Execution Successful ===
```